

Place Value Explorer

Your first number mission

Name: _____

Date: _____

PAGE 1 | WARM-UP

YOUR MISSION Show what you already know about big numbers.

SMART MOVE A drawing or good guess is useful.

Try these on your own. You are not expected to know everything yet.

1. Read each number aloud: 4,706 3,090 8,215

2. Write 6,384 in expanded form.

3. Write 4,099 and 4,900 using words.

4. Place $<$, $>$, or $=$ in the box: 4,099 [] 4,900

5. Round 647 to the nearest 10. Explain with a number line.

6. Round 647 to the nearest 100. Explain with a number line.

Mental Math Moves

Find a clever route through each number.

Name: _____

Date: _____

PAGE 2 | NUMBER MOVES

YOUR MISSION Solve in your head, then leave a trail that shows your idea.

SMART MOVE Friendly numbers can help.

Use words, jumps, or quick sketches. You do not need the vertical method.

1. $399 + 58$

2. $1,000 - 250$

3. $75 + 68$

4. $500 - 198$

5. What is 1 more, 10 more, and 100 more than 3,794?

Which problem felt easiest? Why?

Build It! Place-Value Mat

Turn a number into pieces, digits, and words.

Name: _____

Date: _____

PAGE 3 | REUSABLE MATH TOOL

YOUR MISSION Put each value piece in its matching column.

SMART MOVE Build first. Add digits second.

Use the large boxes for pieces or drawings. Use the small boxes for one digit in each place.

Thousands	Hundreds	Tens	Ones
digit: _____	digit: _____	digit: _____	digit: _____

My number in standard form:

My number in expanded form:

My number in words:

Build It, Say It, Write It

Crack each number's place-value code.

Name: _____

Date: _____

PAGE 4 | GUIDED PRACTICE

YOUR MISSION Name every place and write the expanded form.

SMART MOVE Watch for empty places.

The first one is done as an example. Build the others on your mat before writing.

3,245	3 thousands 2 hundreds 4 tens 5 ones Expanded: $3,000 + 200 + 40 + 5$
4,070	____ thousands ____ hundreds ____ tens ____ ones Expanded: _____
6,308	____ thousands ____ hundreds ____ tens ____ ones Expanded: _____
2,619	____ thousands ____ hundreds ____ tens ____ ones Expanded: _____
5,040	____ thousands ____ hundreds ____ tens ____ ones Expanded: _____
7,503	____ thousands ____ hundreds ____ tens ____ ones Expanded: _____

Number Name Match-Up

One number can wear three different names.

Name: _____

Date: _____

PAGE 5 | INDEPENDENT MISSION

YOUR MISSION Complete all three names for every number.

SMART MOVE Point across each row to check.

Fill every empty box. Then read each completed row aloud.

Standard Form	Expanded Form	Word Form
2,619		
5,040		
	7,000 + 500 + 3	
		eight thousand four hundred sixty-two

Choose one number and explain the value of every digit:

The Place-Value Trade Shop

Change the pieces, but keep the same total.

Name: _____

Date: _____

PAGE 6 | TRADING MISSION

YOUR MISSION Make a fair trade: 1 piece to the left equals 10 pieces to the right.

SMART MOVE Build the trade if unsure.

Decompose means break a number into place-value parts. The usual way uses one digit in each place.

1. 1 thousand = _____ hundreds

2. 1 hundred = _____ tens

3. 1 ten = _____ ones

4. Example: $3,245 = 3 \text{ thousands} + 2 \text{ hundreds} + 4 \text{ tens} + 5 \text{ ones}$

5. $3,245 = 2 \text{ thousands} + \text{_____} \text{ hundreds} + 4 \text{ tens} + 5 \text{ ones}$

6. $4,070 = 3 \text{ thousands} + \text{_____} \text{ hundreds} + 7 \text{ tens}$

Break 8,462 apart in the usual way (8 thousands, 4 hundreds, ...):

Trade one thousand for 10 hundreds. Write the new way:

Why is the total still 8,462?

The 14-Piece Puzzle

How many different numbers can you unlock?

Name: _____ Date: _____

PAGE 7 | PUZZLE MISSION

YOUR MISSION Use exactly 14 pieces to make a four-digit number.

SMART MOVE Keep each column from 0 to 9.

Use 0 to 9 pieces in each column. Add the four piece counts to get 14. The first row is an example.

Thousands	Hundreds	Tens	Ones	Pieces	Number
5	4	3	2	14	5,432

How do you know each answer uses exactly 14 pieces?

What is the greatest number you can make? What is the least four-digit number?

Final Number Mission

Show the most important idea from today.

Name: _____

Date: _____

PAGE 8 | SHOW WHAT YOU KNOW

YOUR MISSION Prove that a digit changes value when it moves places.

SMART MOVE An equation or drawing can help.

Use complete sentences, equations, or drawings.

1. In 5,204, what is the value of the digit 5?

2. In 52, what is the value of the digit 5?

3. Why is the 5 in 5,204 worth 100 times the 5 in 52?

4. Write one question you still have, or one interesting thing you noticed.

☐ I feel secure

☐ I need more practice

☐ I need help

Cut-Apart Digit Cards

Print on cardstock. Cut along the grid lines.

PAGE 9 | REUSABLE MANIPULATIVE

0	1	2	3	4
5	6	7	8	9
0	1	2	3	4
5	6	7	8	9

Cut-Apart Thousands and Hundreds

Cut on the dashed lines. Keep the pieces in labeled bags.

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1,000			
1,000			
1,000	100	100	100
100	100	100	100
100	100	100	100
100	100	100	100

Cut-Apart Tens and Ones

Cut on the dashed lines. Now you have enough pieces to build every lesson number.

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